

Contents

1	Compactness	2
1.1	Sequential Compactness	2

1 Compactness

We will now discuss the equivalence of topological and sequential compactness in the reals and in general metric spaces. Let us view the following theorem:

1.1 Sequential Compactness

Theorem 1.1 (Sequential Bolzano Weierstrass). *Any bounded sequence has a convergent subsequence.*

To prove this fact, we show first that any sequence in $\mathbb{R}$ has a monotonic subsequence.

Proof.

□